11. Radium−226 ($^{226}_{88}Ra$) undergoes α-decay into radon (Rn).

(a) Write a nuclear equation for the decay. (2 marks)

....

....

*(b) Given : mass of a radium nucleus = 226.0254 u
mass of a radon nucleus = 222.0176 u
mass of an α particle = 4.0026 u

Calculate the energy released in the decay in MeV. (2 marks)

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(c) 1 curie (Ci) is defined as the activity of 1 g of radium. The activity of a radium source used in laboratories is about 5 μCi. Estimate the number of radium atoms in this source and hence find its activity expressed in disintegrations per second. The half-life of radium−226 is 1600 years and take the mass of one mole of radium as 226 g. (1 μCi = 1×10^{-6} Ci) (3 marks)

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10. Scientists had been experimenting controlled fusion in a nuclear reactor in which deuterium ($^{2}_{1}$H) and tritium ($^{3}_{1}$H) undergo the following nuclear fusion:

$$^{2}_{1}\text{H} + ^{3}_{1}\text{H} \rightarrow ^{4}_{2}\text{He} + ^{1}_{0}\text{n} + \text{energy released}$$

Given: mass of a deuterium nucleus = 2.014102 u
mass of a tritium nucleus = 3.016049 u
mass of a helium nucleus = 4.002602 u
mass of a neutron = 1.008665 u

*(a) Calculate the energy released, in MeV, in the above nuclear fusion. (2 marks)

(b) A deuterium nucleus and a tritium nucleus have to be within 10^{-15} m for nuclear fusion to occur and that a large amount of work done (about 0.4 MeV) is needed to bring two well separated nuclei to such a close distance.

(i) Explain why a large amount of work done is needed and state the kind of energy this work done has become. (2 marks)

In the nuclear reactor, deuterium and tritium exist as plasma, which is a mixture of ions at a very high temperature.

(ii) Explain why a very high temperature is needed for nuclear fusion to occur. (1 mark)

*(iii)Estimate the order of magnitude of the minimum temperature at which fusion of deuterium and tritium nuclei would be possible if the plasma can be regarded as an ideal gas. (Hint: For an ideal gas, the gas molecules each is assumed to have an average kinetic energy $E_K = \frac{3RT}{2N_A}$) (2 marks)

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9. Part of the decay series of uranium-238 (U-238) is shown below. The end product lead-206 (Pb-206) is stable.

$$^{238}_{92}\mathrm{U} \xrightarrow[4.5\times10^9 \text{ years}]{\alpha} \mathrm{Th} \xrightarrow[21 \text{ days}]{\beta} \mathrm{Pa} \rightarrow \ldots \rightarrow {}^{206}_{82}\mathrm{Pb}$$

(a) When a U-238 nucleus decays to a Pb-206 nucleus, how many α-particle(s) and β-particle(s) are emitted ? (2 marks)

(b) As the first decay in the above chain from U to Th has a half-life much longer than those of subsequent decays, the decay from U-238 to Pb-206 can be simplified to *a single decay* with half-life 4.5×10^9 years:

$$^{238}_{92}\mathrm{U} \xrightarrow[4.5\times10^9 \text{ years}]{} {}^{206}_{82}\mathrm{Pb}$$

Suppose that a uranium-bearing rock contains only U-238 and no Pb-206 at the time when it was formed long ago by solidification of molten material. In a particular sample of the rock, it is found that the ratio $\dfrac{\text{number of Pb - 206 atoms}}{\text{number of U - 238 atoms}}$ is $\dfrac{2}{3}$ *at present*.

(i) Estimate the age of the rock. Assume that all Pb-206 atoms come from the decay of U-238 originally present in the sample and ignore the small number of U-238 atoms which have decayed but have not yet become Pb-206. (2 marks)

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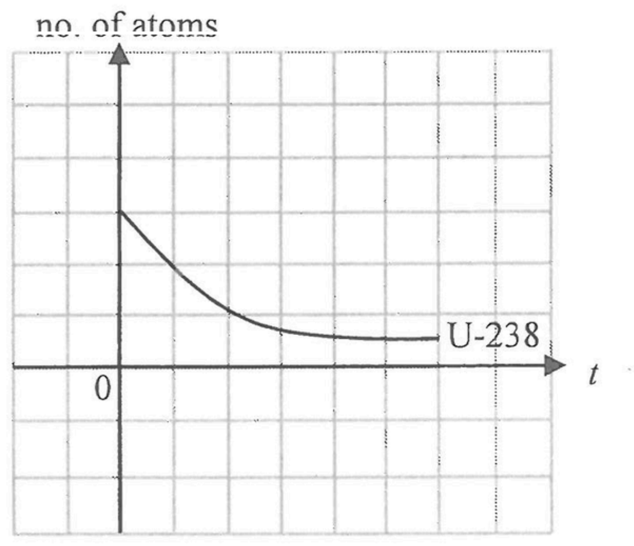
(ii) State, with a reason, whether the answer in (b)(i) is an overestimate or an underestimate of the age of the rock if some Pb-206 atoms have actually been lost. (2 marks)

(iii) The graph in Figure 9.1 shows how the number of U-238 atoms in the sample varies with time t subsequently while $t = 0$ denotes *the present time*. On Figure 9.1, sketch a graph to show the variation of the number of Pb-206 atoms in the sample with time. (2 marks)

Figure 9.1



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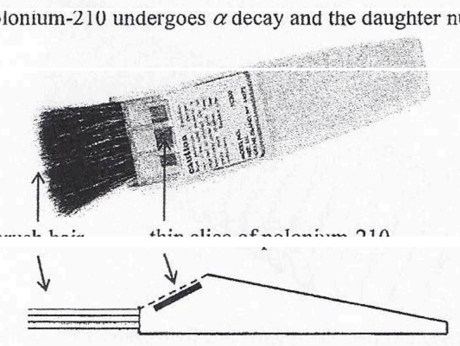
10. Dust may adhere to the surfaces of photos and films due to electrostatic attraction. To remove the dust effectively, a special brush with a thin slice of polonium-210 ($^{210}_{84}\text{Po}$) fixed near the brush hair as shown in Figure 10.1 may be used. Polonium-210 undergoes α decay and the daughter nucleus lead (Pb) is stable.

Figure 10.1



(a) Write a nuclear equation for the decay of polonium-210. (2 marks)

(b) Briefly explain how the α particles help clean the charged dust. (2 marks)

(c) Briefly explain why the polonium-210 slice must be fixed near to the brush hair. (1 mark)

*(d) The manufacturer recommends that the brush should be returned to the factory for replacement of the polonium-210 slice every year. Taking the activity of a newly replaced polonium-210 slice as 1 unit, find its activity after one year (365 days). Given: half-life of polonium-210 is 138 days. (2 marks)

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List of data, formulae and relationships

Data

molar gas constant	$R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$
Avogadro constant	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
acceleration due to gravity	$g = 9.81 \text{ m s}^{-2}$ (close to the Earth)
universal gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
speed of light in vacuum	$c = 3.00 \times 10^8 \text{ m s}^{-1}$
charge of electron	$e = 1.60 \times 10^{-19} \text{ C}$
electron rest mass	$m_e = 9.11 \times 10^{-31} \text{ kg}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$
permeability of free space	$\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$
atomic mass unit	$u = 1.661 \times 10^{-27} \text{ kg}$ (1 u is equivalent to 931 MeV)
astronomical unit	$AU = 1.50 \times 10^{11} \text{ m}$
light year	$ly = 9.46 \times 10^{15} \text{ m}$
parsec	$pc = 3.09 \times 10^{16} \text{ m} = 3.26 \text{ ly} = 206265 \text{ AU}$
Stefan constant	$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
Planck constant	$h = 6.63 \times 10^{-34} \text{ J s}$

Rectilinear motion

For uniformly accelerated motion :

$$v = u + at$$
$$s = ut + \frac{1}{2}at^2$$
$$v^2 = u^2 + 2as$$

Mathematics

Equation of a straight line $y = mx + c$

Arc length $= r\theta$

Surface area of cylinder $= 2\pi rh + 2\pi r^2$

Volume of cylinder $= \pi r^2 h$

Surface area of sphere $= 4\pi r^2$

Volume of sphere $= \frac{4}{3}\pi r^3$

For small angles, $\sin\theta \approx \tan\theta \approx \theta$ (in radians)

Astronomy and Space Science		Energy and Use of Energy	
$U = -\frac{GMm}{r}$	gravitational potential energy	$E = \frac{\Phi}{A}$	illuminance
$P = \sigma AT^4$	Stefan's law	$\frac{Q}{t} = \kappa\frac{A(T_H - T_C)}{d}$	rate of energy transfer by conduction
$\left\|\frac{\Delta f}{f_0}\right\| \approx \frac{v}{c} \approx \left\|\frac{\Delta\lambda}{\lambda_0}\right\|$	Doppler effect	$U = \frac{\kappa}{d}$	thermal transmittance U-value
		$P = \frac{1}{2}\rho Av^3$	maximum power by wind turbine
Atomic World		**Medical Physics**	
$\frac{1}{2}m_e {v_{max}}^2 = hf - \phi$	Einstein's photoelectric equation	$\theta \approx \frac{1.22\lambda}{d}$	Rayleigh criterion (resolving power)
$E_n = -\frac{1}{n^2}\left\{\frac{m_e e^4}{8h^2\varepsilon_0^2}\right\} = -\frac{13.6}{n^2} \text{ eV}$	energy level equation for hydrogen atom	$\text{power} = \frac{1}{f}$	power of a lens
		$L = 10\log\frac{I}{I_0}$	intensity level (dB)
$\lambda = \frac{h}{p} = \frac{h}{mv}$	de Broglie formula	$Z = \rho c$	acoustic impedance
$\theta \approx \frac{1.22\lambda}{d}$	Rayleigh criterion (resolving power)	$\alpha = \frac{I_r}{I_0} = \frac{(Z_2 - Z_1)^2}{(Z_2 + Z_1)^2}$	intensity reflection coefficient
		$I = I_0 e^{-\mu x}$	transmitted intensity through a medium

A1.	$E = mc\,\Delta T$	energy transfer during heating and cooling	D1.	$F = \frac{Q_1Q_2}{4\pi\varepsilon_0 r^2}$	Coulomb's law
A2.	$E = l\,\Delta m$	energy transfer during change of state	D2.	$E = \frac{Q}{4\pi\varepsilon_0 r^2}$	electric field strength due to a point charge
A3.	$pV = nRT$	equation of state for an ideal gas	D3.	$E = \frac{V}{d}$	electric field between parallel plates (numerically)
A4.	$pV = \frac{1}{3}Nm\overline{c^2}$	kinetic theory equation	D4.	$R = \frac{\rho l}{A}$	resistance and resistivity
A5.	$E_K = \frac{3RT}{2N_A}$	molecular kinetic energy	D5.	$R = R_1 + R_2$	resistors in series
			D6.	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$	resistors in parallel
B1.	$F = m\frac{\Delta v}{\Delta t} = \frac{\Delta p}{\Delta t}$	force	D7.	$P = IV = I^2R$	power in a circuit
B2.	$\text{moment} = F \times d$	moment of a force	D8.	$F = BQv\sin\theta$	force on a moving charge in a magnetic field
B3.	$E_P = mgh$	gravitational potential energy	D9.	$F = BIl\sin\theta$	force on a current-carrying conductor in a magnetic field
B4.	$E_K = \frac{1}{2}mv^2$	kinetic energy	D10.	$B = \frac{\mu_0 I}{2\pi r}$	magnetic field due to a long straight wire
B5.	$P = Fv$	mechanical power	D11.	$B = \frac{\mu_0 NI}{l}$	magnetic field inside a long solenoid
B6.	$a = \frac{v^2}{r} = \omega^2 r$	centripetal acceleration	D12.	$\varepsilon = N\frac{\Delta\Phi}{\Delta t}$	induced e.m.f.
B7.	$F = \frac{Gm_1m_2}{r^2}$	Newton's law of gravitation	D13.	$\frac{V_s}{V_p} \approx \frac{N_s}{N_p}$	ratio of secondary voltage to primary voltage in a transformer
C1.	$\Delta y = \frac{\lambda D}{a}$	fringe width in double-slit interference	E1.	$N = N_0 e^{-kt}$	law of radioactive decay
C2.	$d\sin\theta = n\lambda$	diffraction grating equation	E2.	$t_{\frac{1}{2}} = \frac{\ln 2}{k}$	half-life and decay constant
C3.	$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$	equation for a single lens	E3.	$A = kN$	activity and the number of undecayed nuclei
			E4.	$\Delta E = \Delta mc^2$	mass-energy relationship

10. (a) The equation below represents nuclear fission of uranium-235 (U-235).

$$^{235}_{92}\text{U} + {}^{1}_{0}\text{n} \rightarrow {}^{141}_{56}\text{Ba} + {}^{92}_{36}\text{Kr} + x\,{}^{1}_{0}\text{n} + 200\text{ MeV}$$

(i) What is the value of x ? (1 mark)

(ii) State a necessary condition for chain reaction of fission to occur. (1 mark)

Scientists found evidence in Oklo, Africa that natural nuclear fission occurred two billion (2×10^9) years ago. The uranium mineral ore mined from Oklo **at present** is found to have 0.6% concentration by mass of U-235 (see the table below), which is much lower than usual.

(b) The table gives the information of U-235 and U-238 in a sample of uranium mineral ore found in Oklo. Given: half-life of U-235 = 7.04×10^8 years

	2×10^9 years ago	**at present**
U-235	m_0 kg	0.060 kg (i.e. 0.6% concentration by mass)
U-238	13.556 kg	9.940 kg (i.e. 99.4% concentration by mass)

*(i) Estimate the amount m_0 (in kg) of U-235 in the sample 2×10^9 years ago. (2 marks)

(ii) Hence determine whether natural nuclear fission of U-235 was possible 2×10^9 years ago. For fission of U-235 to happen, its concentration by mass in the uranium mineral ore has to be at least 3%. (1 mark)

There must be underground water in the vicinity of this uranium-rich mineral deposit for natural nuclear fission to be possible. Since water can slow down the fast neutrons from fission, these neutrons can easily be captured by U-235.

(c) In fact the chain reaction stopped even before the concentration by mass of U-235 dropped to 3%. Explain why this occurred. (2 marks)

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10. Given: mass of proton $= 1.0073$ u
mass of α particle $= 4.0015$ u
mass of ${}^{14}_{7}\text{N}$ nucleus $= 13.9993$ u
mass of ${}^{17}_{8}\text{O}$ nucleus $= 16.9947$ u

When a stationary ${}^{14}_{7}\text{N}$ nucleus is bombarded by an α particle, the following nuclear reaction can be triggered with products ${}^{17}_{8}\text{O}$ and X fly off:

$$\alpha + {}^{14}_{7}\text{N} \longrightarrow {}^{17}_{8}\text{O} + X$$

(a) What is X? (1 mark)

*(b) Based on energy consideration, estimate the minimum kinetic energy, in MeV, of the α particle required for such a nuclear reaction to occur. (2 marks)

(c) However, when conservation of momentum is also taken into account, the α particle must possess a kinetic energy greater than that found in (b) to bring about such a reaction. Explain. (2 marks)

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10. Deuterium (^2_1H) and tritium (^3_1H) are isotopes of hydrogen. The fusion of two deuterium nuclides can be represented by the following equation:

$$^2_1H + ^2_1H \longrightarrow ^3_2He + ^1_0n$$

Given: mass of ^2_1H = 2.014102 u
mass of ^3_2He = 3.016029 u
mass of ^1_0n = 1.008665 u

*(a) 1 in 6420 atoms of hydrogen in nature is a deuterium. Estimate the maximum energy, in MeV, that could be produced from this fusion reaction with 1 mole of hydrogen nuclides. (3 marks)

(b) If conditions are altered, the fusion of two deuterium nuclides may **not** produce a helium (He) nucleus. Complete the equation below for such a possible fusion reaction. (1 mark)

$$^2_1H + ^2_1H \longrightarrow \text{.........} + \text{.........}$$

(c) Both fission and fusion produce energy. State **TWO** advantages of using fusion as an energy source over fission. (2 marks)

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9. Part of the decay series of thorium-232 (Th-232) is shown below. Th-232 decays to radium-228 (Ra-228) with a half-life of 1.41×10^{10} years. The end product lead-208 (Pb-208) is stable.

$$^{232}_{90}\mathrm{Th} \longrightarrow {}^{228}_{88}\mathrm{Ra} \xrightarrow{\beta} \mathrm{Ac} \rightarrow \ldots \rightarrow {}^{208}_{82}\mathrm{Pb}$$

(a) (i) State the kind of radiation emitted by Th-232. (1 mark)

(ii) How many α- and β- decay(s) has occurred before a Th-232 nucleus becomes a Pb-208 nucleus ? (2 marks)

*(b) Estimate the proportion of Th-232 decaying in 10 years. Give your answer correct to 2 significant figures. (2 marks)

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(c) By adding thorium oxide (a compound containing Th-232) to glass, thoriated glass having a higher refractive index is produced. Camera lenses manufactured before 1970 are often made from thoriated glass.

Figure 9.1

(i) Although thoriated glass is radioactive, practically it is safe to use a camera with such lenses like the one shown in Figure 9.1. Why ? (1 mark)

(ii) According to the decay series of thorium-232, thorium oxide would end up as lead oxide which is black in colour. Why is there no need to worry that a thoriated glass lens would turn black when used for several years ? (1 mark)

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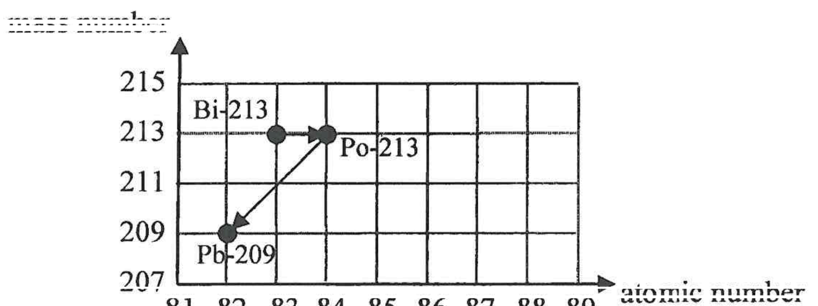
12. (a) Bismuth-213 (Bi-213) is a radionuclide currently undergoing clinical trials for cancer treatment. It is prepared in liquid and injected into the human body, where it targets and destroys cancer cells. Figure 12.1 shows a decay series of Bi-213.

Figure 12.1



(i) Write the nuclear equations corresponding to each of these two decay processes. (2 marks)

*(ii) The half-life of Bi-213 is 46 minutes. At time $t = 0$ s, its number of undecayed nuclei is 8.0×10^{11}. Find its activity (in Bq) at $t = 0$ s. (2 marks)

*(iii) Find the percentage decrease in the activity after 200 minutes. (2 marks)

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(b) In hospital, technicians wear protective gloves to handle liquid radioactive materials, such as those containing Bi-213. To monitor radiation exposure, they wear a specialised ring on their finger, as shown in Figure 12.2(a). Each ring contains a small chip in a plastic enclosure (Figure 12.2(b)), which detects and records radiation doses. The rings are collected every three months to monitor the radiation exposure of the technicians.


Figure 12.2(a)

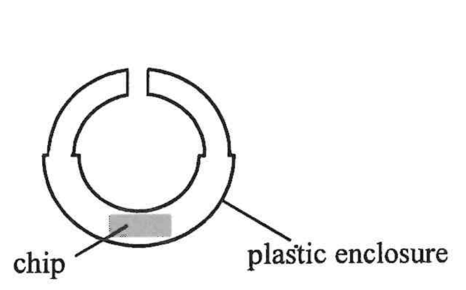

Figure 12.2(b)

(i) Compared to placing it in a badge worn on the chest, what is the advantage of placing the chip in the ring worn on the hand ? (1 mark)

(ii) The ring should be worn under the gloves. Explain briefly. (1 mark)

(iii) Which type(s) of radiation (α, β or γ) can reach the chip of the ring ? (1 mark)

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